Course Code: CSE422
Course Title: Artificial Intelligence
Project Report

Project Title: *Client Term Deposit Subscription Prediction using Machine Learning*

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Introduction

In today's competitive banking and financial services landscape, institutions increasingly rely on data-driven strategies to optimize marketing campaigns and maximize return on investment. One challenge is identifying which customers are most likely to subscribe to specific financial products, such as term deposits. Traditionally, telemarketing campaigns rely on broad, indiscriminate cold-calling, which is resource-intensive, inefficient, and often leads to poor customer experiences due to irrelevant targeting. To address these challenges, this project builds a machine learning based classification system that predicts customer subscription behavior using historical marketing data.

The primary objective is to develop a predictive model that estimates the probability of a customer subscribing to a term deposit. The model is trained based on the customer's personal details such as age, job and marital status, and their financial history which includes housing, loans and credit, along with economic indicators like euribor rates and consumer price index to answer a simple Yes or No question. By learning from these attributes, the model can help bank marketing teams prioritize high-value leads, reduce operational costs by filtering our non-responders and significantly improve conversion rates through targeted engagement.

The motivation behind this project stems from the practical need for smarter, more efficient allocation of marketing resources. Automation using this model not only enhances operational efficiency but also reduces intrusion on customers who are unlikely to be interested.

In this project, we explore and compare three distinct classification algorithms: Logistic Regression, Decision Tree Classifier and Neural Network (MLP). We evaluate their performance using comprehensive metrics, including Prediction Accuracy, Precision and Recall comparison, F1 score, Confusion Matrix, and the ROC-AUC score in order to identify the most suitable model for accurate and reliable term deposit prediction.

Dataset Description

We use a structured dataset containing information on 41188 individuals collected by the Bank Marketing team, each described by several features that may influence the likelihood of that individual subscribing to a term deposit. Each individual is labeled with the answer to the question of whether they will subscribe or not (1 for Yes, 0 for No), which is the target variable for prediction. The dataset has 41188 distinct datapoints and 21 features, which are the following:

Quantitative Features:

There are 10 quantitative features:

- **age** – age of the client
- **duration** – duration of last contact (in seconds)
- **campaign** – number of contacts during the current campaign
- **pdays** – number of days since last contact from a previous campaign
- **previous** – number of contacts before this campaign
- **emp.var.rate** – employment variation rate
- **cons.price.idx** – consumer price index
- **cons.conf.idx** – consumer confidence index
- **euribor3m** – euribor 3-month rate
- **nr.employed** – number of employees

Categorical Features:

There are 10 categorical features:

- **job** – type of job
- **marital** – marital status
- **education** – education level
- **default** – has credit in default?
- **housing** – has housing loan?
- **loan** – has personal loan?
- **contact** – contact communication type
- **month** – last contact month
- **day_of_week** – last contact day
- **poutcome** – outcome of previous campaign

This dataset represents a Classification problem as the target variable 'y' consists of two distinct categories (Yes and No). Since the output for this dataset is categorical rather than continuous, classification models are more suitable for this dataset than a regression model.

We have to encode the Categorical variables as the dataset contains a significant number of categorical features such as job, marital, education etc., which creates a problem as most machine learning models use numerical values and cannot understand the text based categories directly. So we encoded these variables into numerical form before training the models using One-Hot encoding for variables like job, marital, contact, etc. and Label encoding for other variables like default, housing and loan.

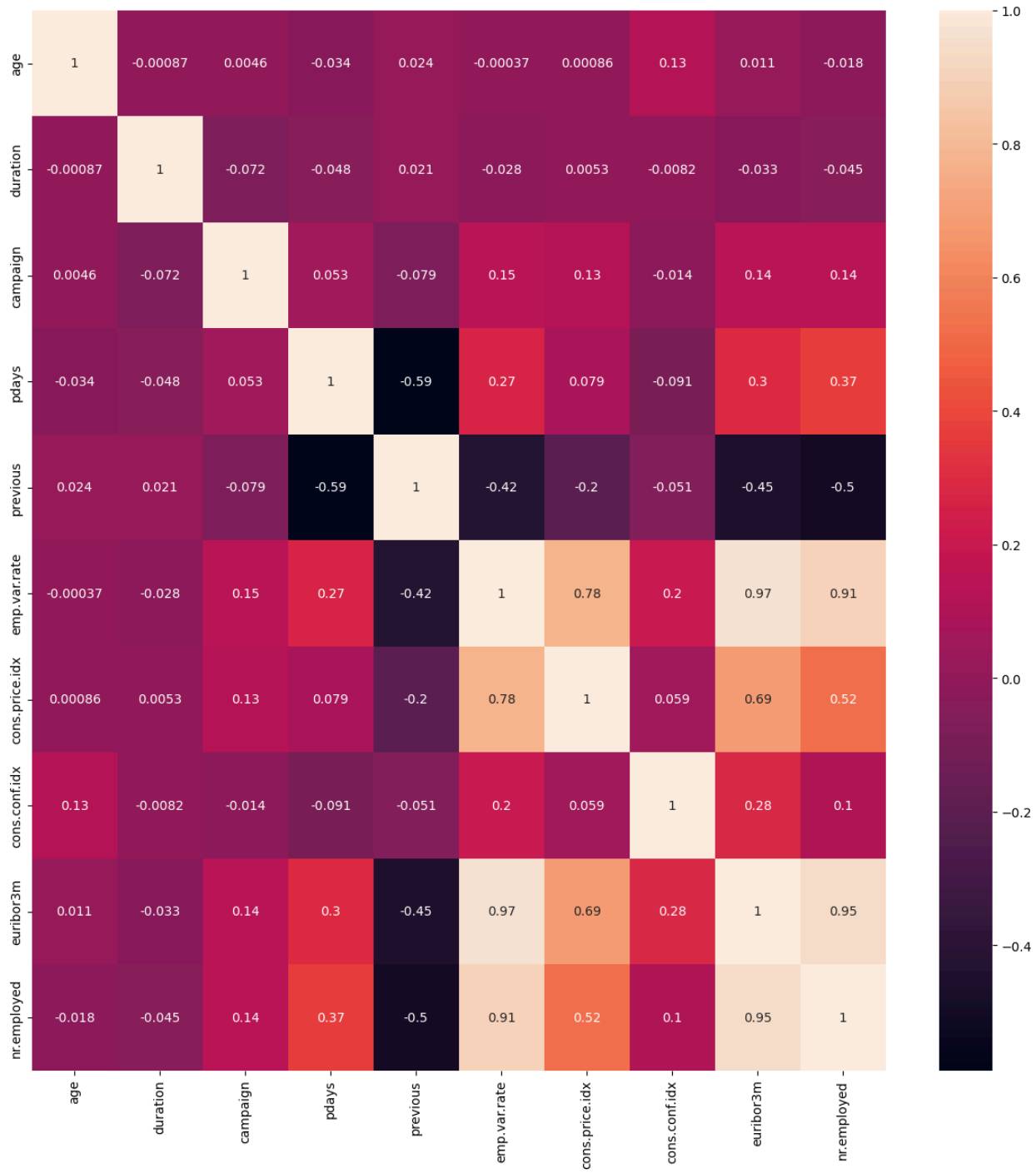
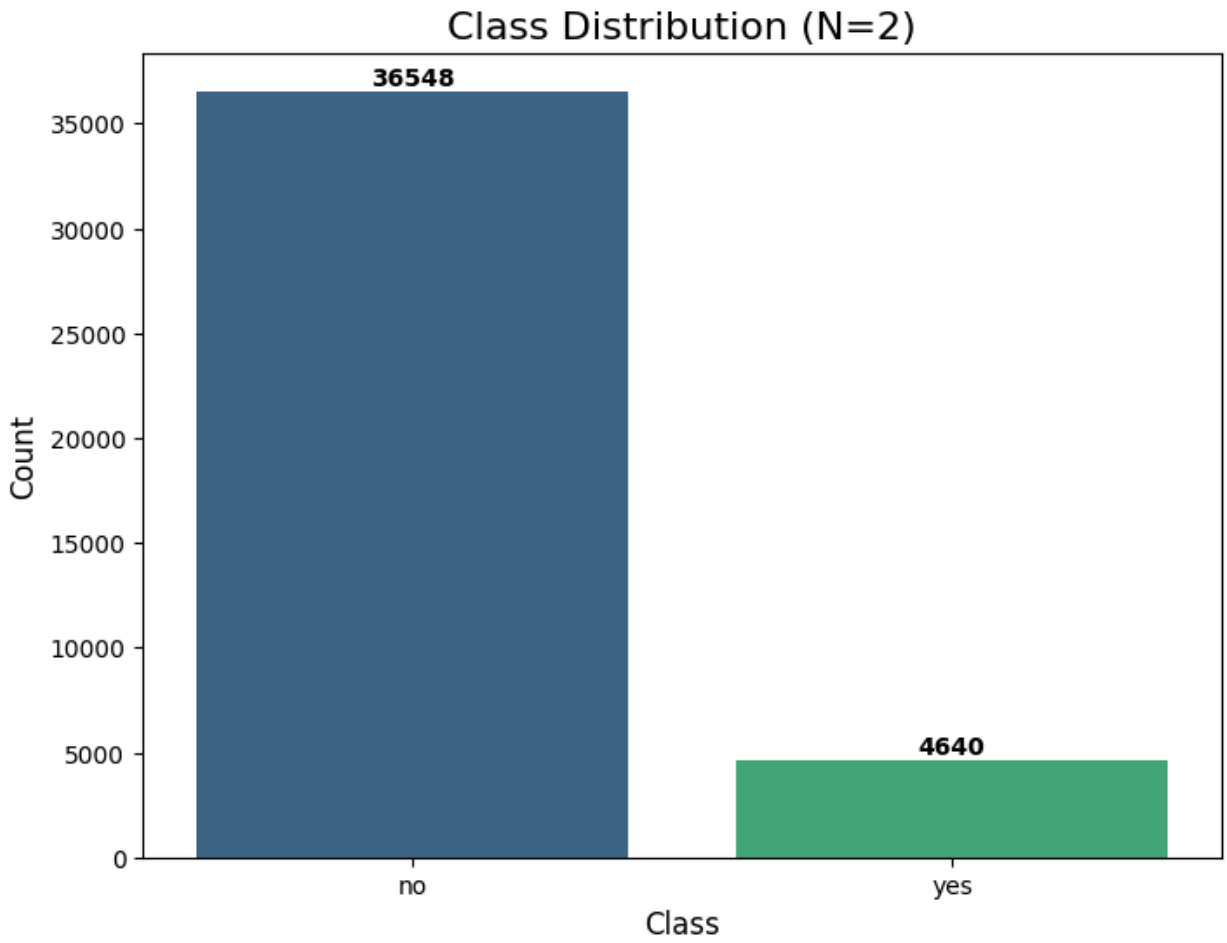


Fig 1. : Correlation of all the features using a heatmap

The dataset appears to be an Imbalanced dataset as in the output feature, one unique class has a substantially larger number of instances than the other (Yes 11.27%, No 88.73%)



Correlation analysis reveals the direction and strength of relationships in the dataset, identifying key predictors like call duration (positive impact) and economic factors like nr.employed (negative impact) on customer subscriptions. These insights help streamline the model by highlighting the most influential features while identifying redundant features to remove.

Dataset Pre-processing

- **Null / Missing Values**
  We have the following count of null values in our data:

```
df.isnull().sum()
```

	0
age	0
job	0
marital	0
education	0
default	0
housing	0
loan	0
contact	0
month	0
day_of_week	0
duration	0
campaign	0
pdays	0
previous	0
poutcome	0
emp.var.rate	0
cons.price.idx	0
cons.conf.idx	0
euribor3m	0
nr.employed	0
y	0

dtype: int64

Dropping Unnecessary Features

We dropped the feature "duration" as it artificially increases the accuracy of the model. Since using the call duration, which is the strongest predictor in the correlation heatmap, to train the model we are essentially giving out the answer, which in a real-world scenario would not exist. Thus our model appears to perform better than it actually is if we use that data.

Feature Scaling

We used **Standardization** (specifically StandardScaler from Scikit-Learn) to scale the features.

We did this by:

1. We calculated the average (mean) and spread (standard deviation) for every column using **only the training data**.
2. We subtracted the mean and divided by the standard deviation for every single value.
3. ***Formula:*** $z = \frac{x - \mu}{\sigma}$
3. We applied this same math to the **test data** using the training data's stats (so we didn't "cheat" by peeking at the test data's distribution).

All the features (like age which is 20-80 and euribor3m which is 0-5) were converted to the same scale (centered around 0), so the model treats them equally.

Encoding

We used two distinct encoding methods to translate text data into numbers that the models could understand:

1. **Label Encoding (Target Variable y):**

   We converted the "Yes/No" answers into 1 and 0. Since the target is binary, so a simple 0/1 binary is the most efficient way to represent the outcome.
2. **One-Hot Encoding (Features x):**

   We used pd.get_dummies() to explode categorical columns like job into separate columns for every option. This prevents the model from assuming a false hierarchy. It treats every category as equally distinct.

Train-Test Split

We split the processed data into a training set (80%) and a test set (20%) for this project.

Model Training

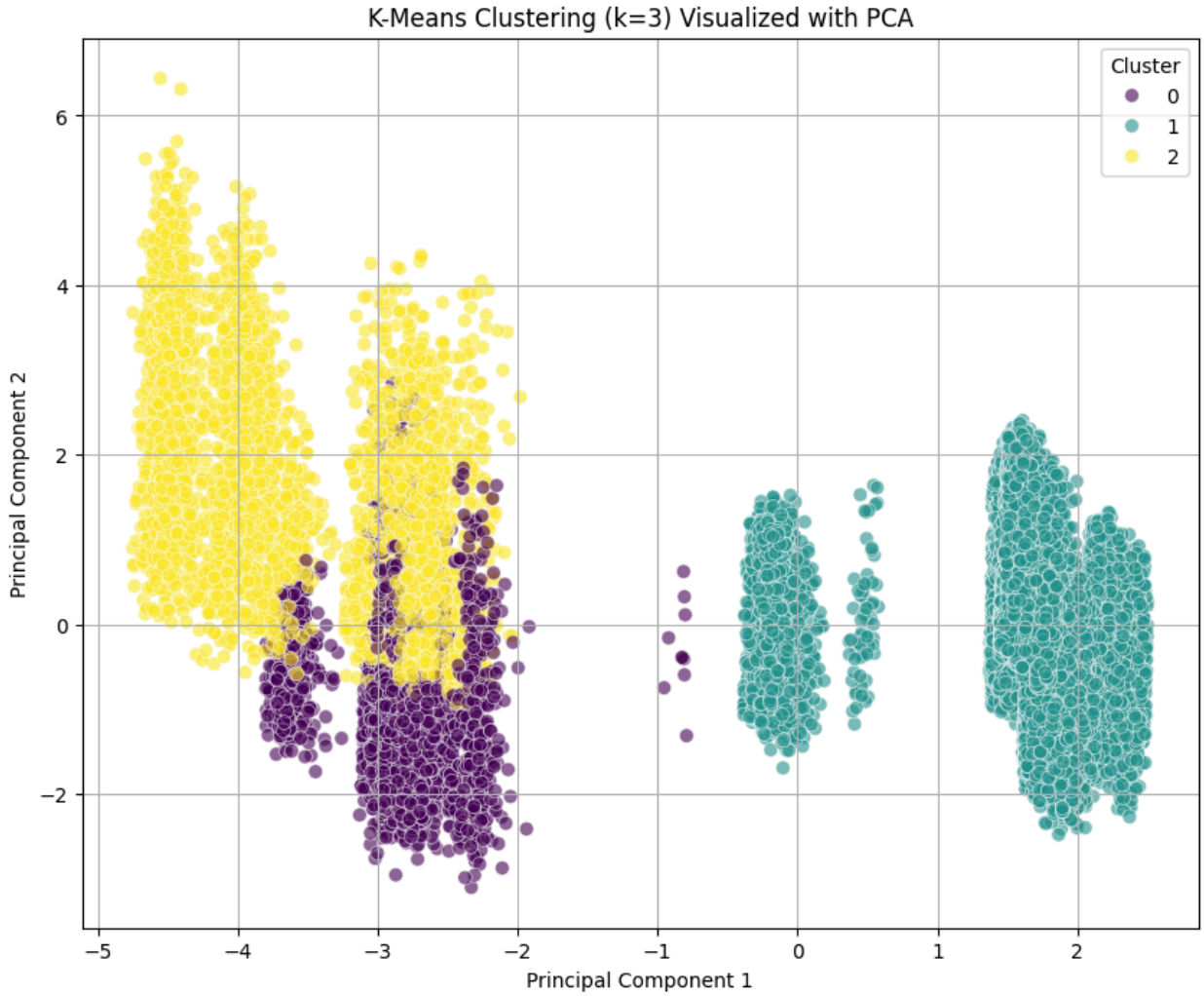
We trained three different classification models on the prepared dataset: Logistic Regression, Decision Tree and Neural Network. We chose these to cover a range from simple linear to more complex non linear learners. All models were trained on the same training data and evaluated on the test set using accuracy (percentage of correct predictions) as the metric.

Logistic Regression: We chose logistic regression for model training because it is specifically designed for a binary classification problem similar to our dataset and models the probability of an outcome using a logistical function (sigmoid). It can also efficiently handle our dataset which is a mix of numerical and encoded categorical features and performs well on similarly large tabular datasets.

Decision Tree Classifier: A decision tree functions by learning through rule based decisions by splitting data on feature values. It can also capture the non-linear relationships between features and the target variable making the dataset easy to interpret and visualize. It is suitable for our dataset as it handles both numerical and encoded categorical values without any feature scaling and helps us understand the non-linear features that influence an individual's decision behaviour.

Neural Network: We included a neural network because it’s a powerful model capable of automatically learning complex feature combinations and non-linear patterns. It is very efficient in advanced predictive modeling. It can handle large datasets with ease and can model complex decision-making behaviour of individual clients, producing higher predictive performance than most traditional models.

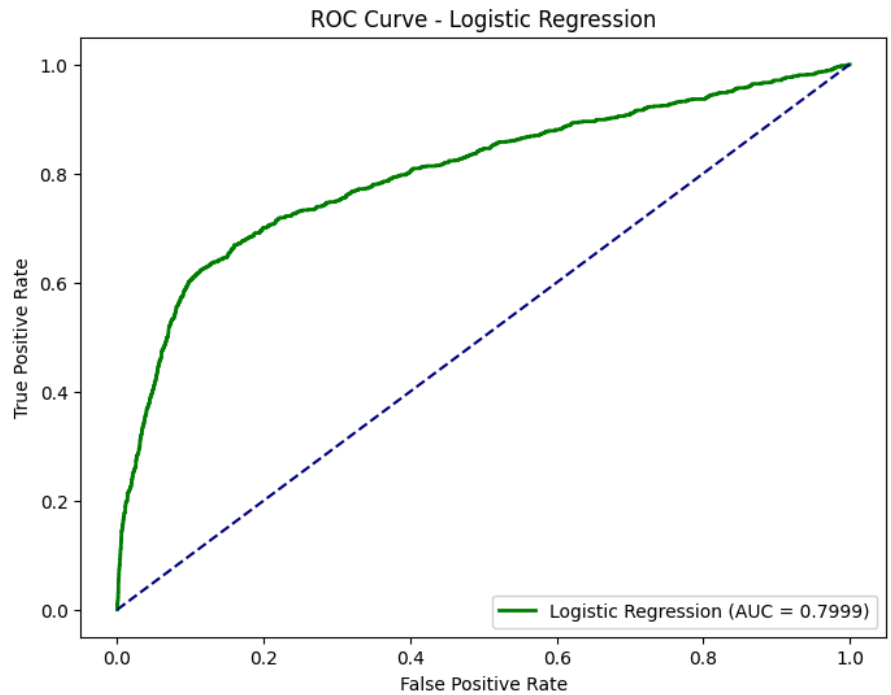
K-Means Clustering



Model Evaluation and Comparison Analysis

Logistic Regression

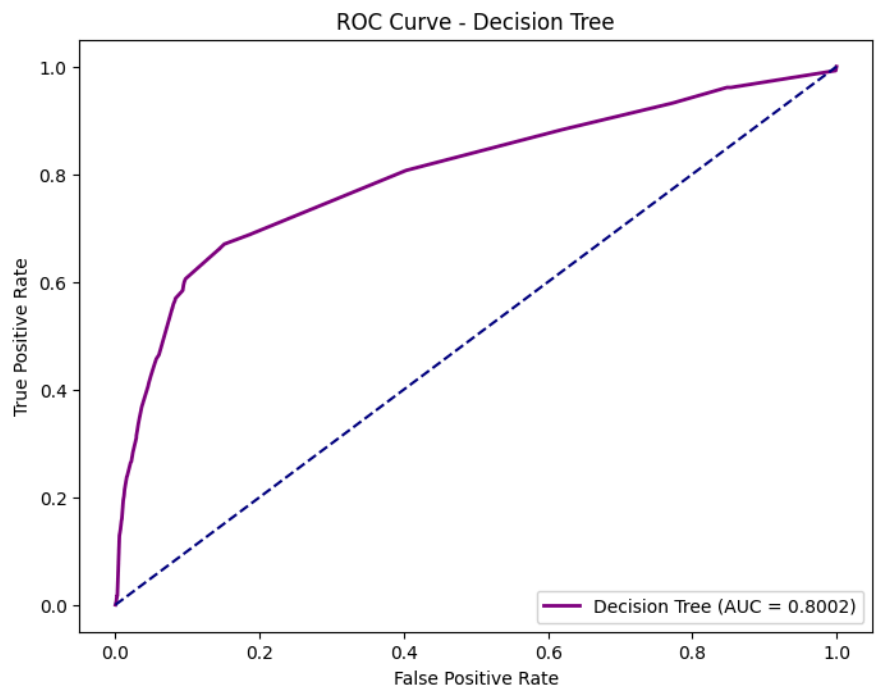
- **Prediction Accuracy:** 89.79%
- **Precision:** 65% for Yes (1) and 91% for No (0)
- **Recall:** 20% for Yes (1) and 99% for No (0)
- **F1-Score:** 0.31 for Yes (1) and 0.94 for No (0)
- **AUC Score:** 0.7999




```
Model Accuracy: 0.897887323943662

Classification Report:
              precision    recall  f1-score   support

           0       0.91      0.99      0.94      7308
           1       0.65      0.20      0.31       928

    accuracy                           0.90      8236
   macro avg       0.78      0.59      0.63      8236
weighted avg       0.88      0.90      0.87      8236
```

Decision Tree

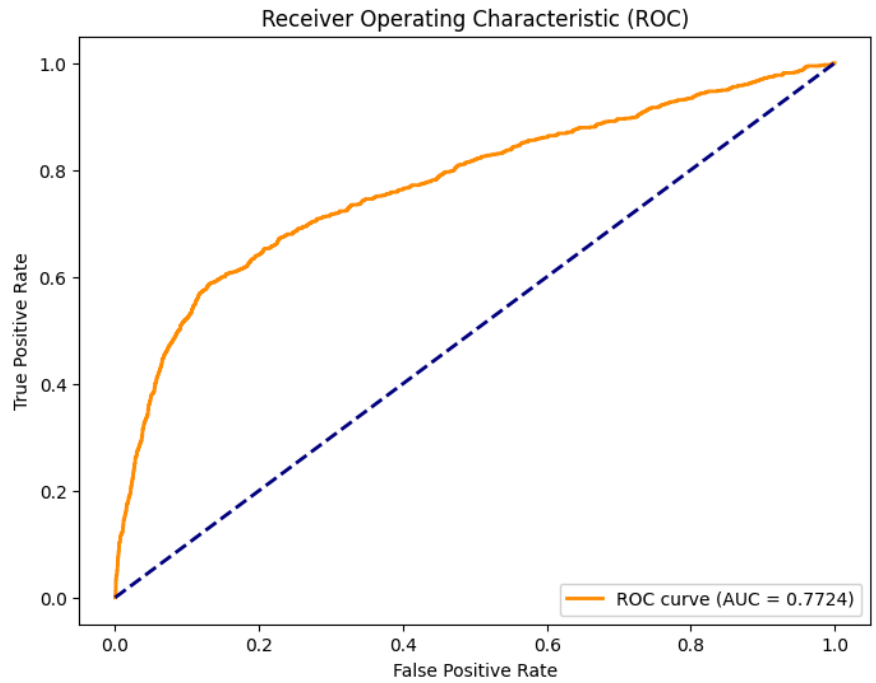
- **Accuracy:** 89.98%
- **Precision:** 65% for Yes (1) and 91% for No (0)
- **Recall:** 24% for Yes (1) and 98% for No (0)
- **F1-Score:** 0.35 for Yes (1) and 0.95 for No (0)
- **AUC Score:** 0.8002




```
Decision Tree Accuracy: 0.8998300145701797

Classification Report:
              precision    recall  f1-score   support

           0       0.91      0.98      0.95      7308
           1       0.65      0.24      0.35       928

    accuracy                           0.90      8236
   macro avg       0.78      0.61      0.65      8236
weighted avg       0.88      0.90      0.88      8236
```

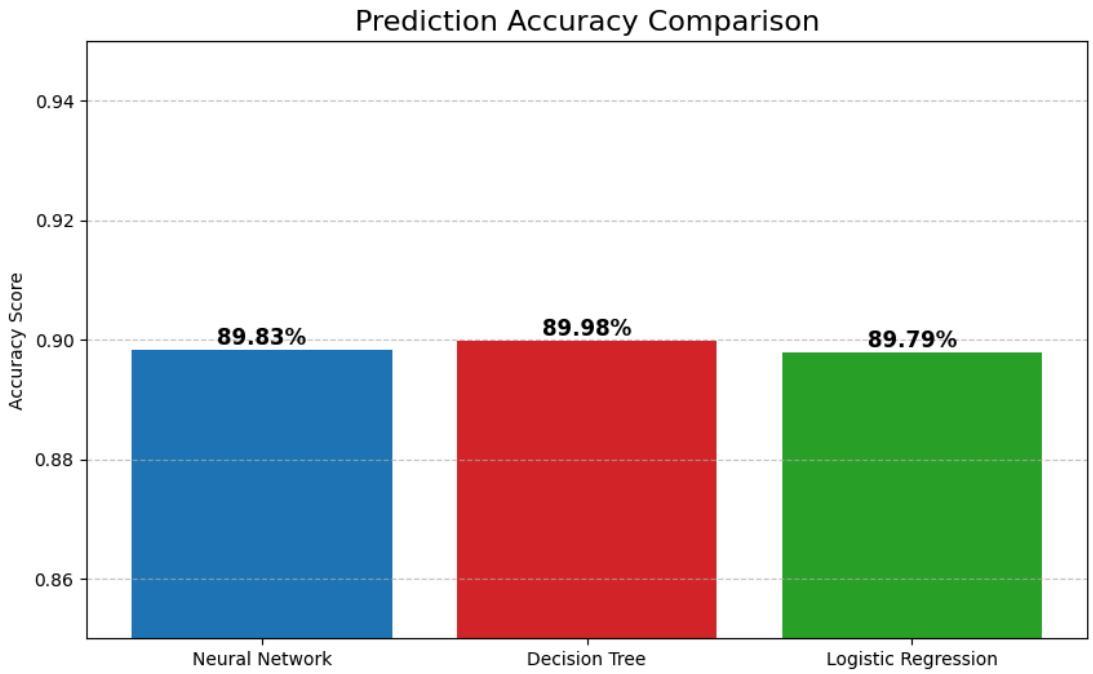
Neural Network

- **Accuracy:** 89.82%
- **Precision:** 0.65% for Yes (1) and 0.91% for No (0)
- **Recall:** 0.21% for Yes (1) and 0.99% for No (0)
- **F1-Score:** 0.32 for Yes (1) and 0.94 for No (0)
- **AUC score:** 0.7724

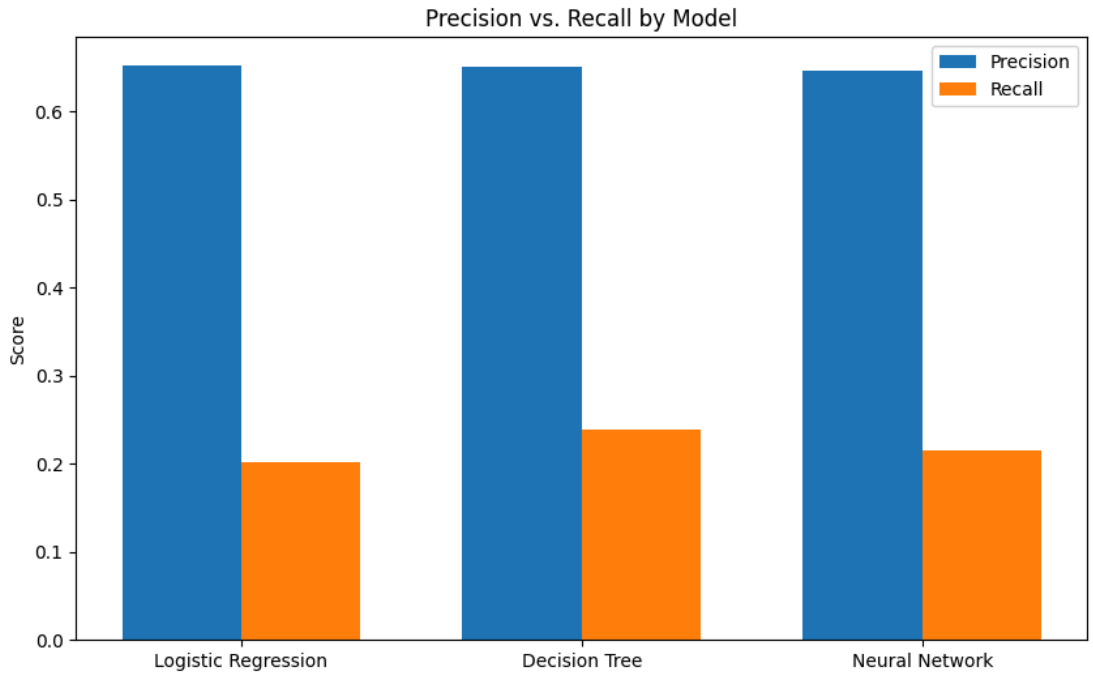



```
Neural Network Accuracy: 0.898251578436134

Classification Report:
              precision    recall  f1-score   support

           0       0.91      0.99      0.94      7308
           1       0.65      0.21      0.32       928

    accuracy                           0.90      8236
   macro avg       0.78      0.60      0.63      8236
weighted avg       0.88      0.90      0.87      8236
```

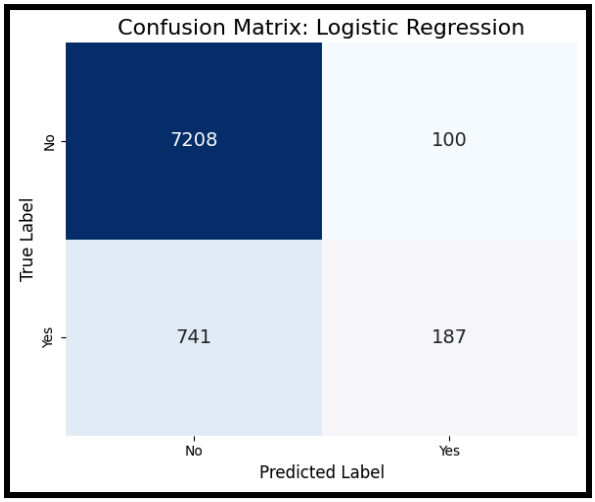
Prediction Accuracy Comparison by Model

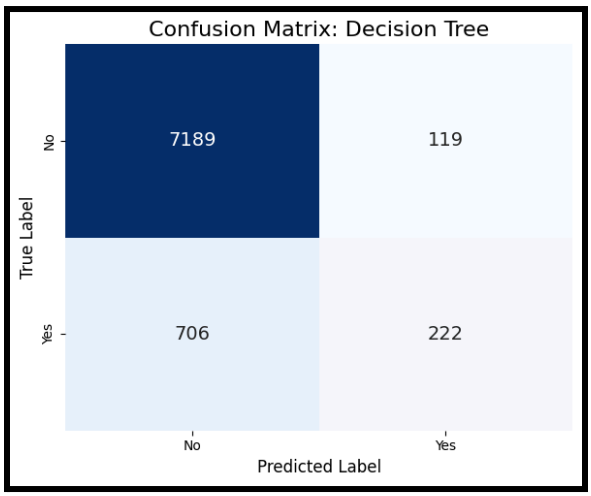


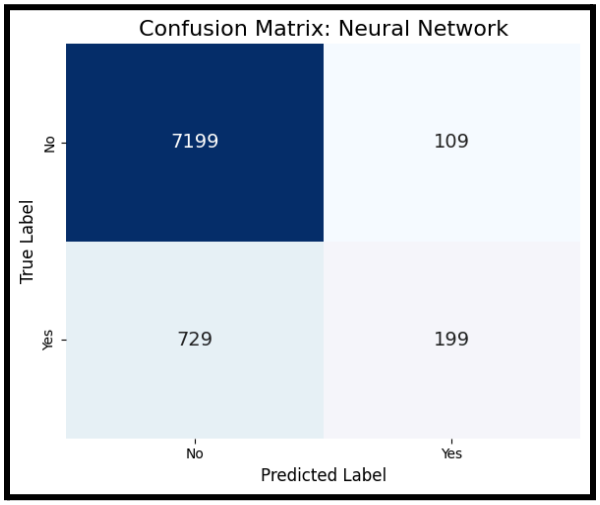
Precision Vs Recall Comparison by Model



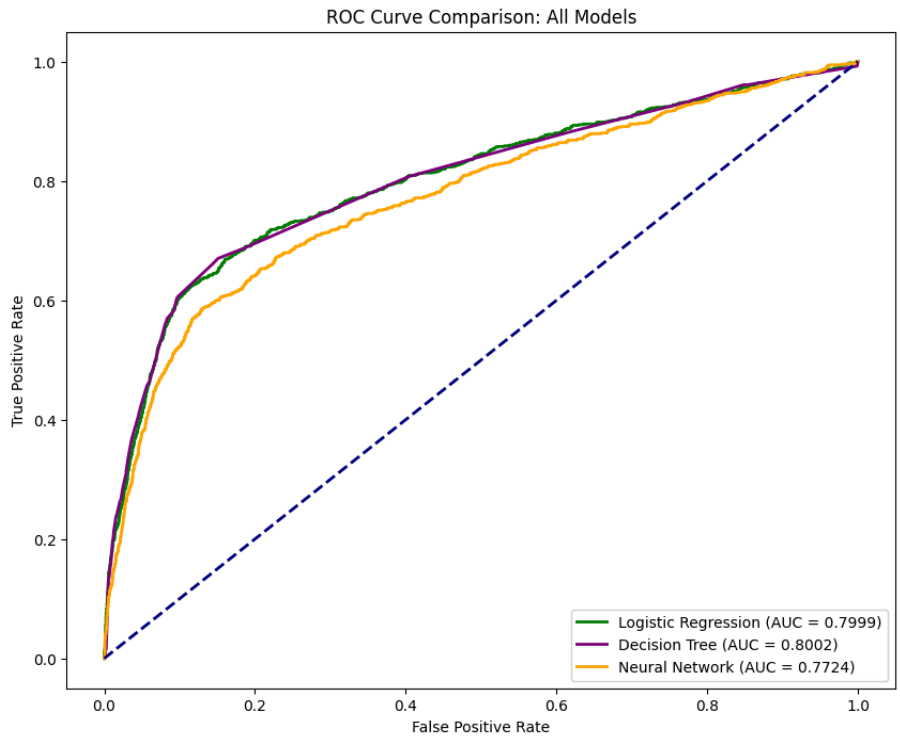
Confusion Matrix Comparison by Model



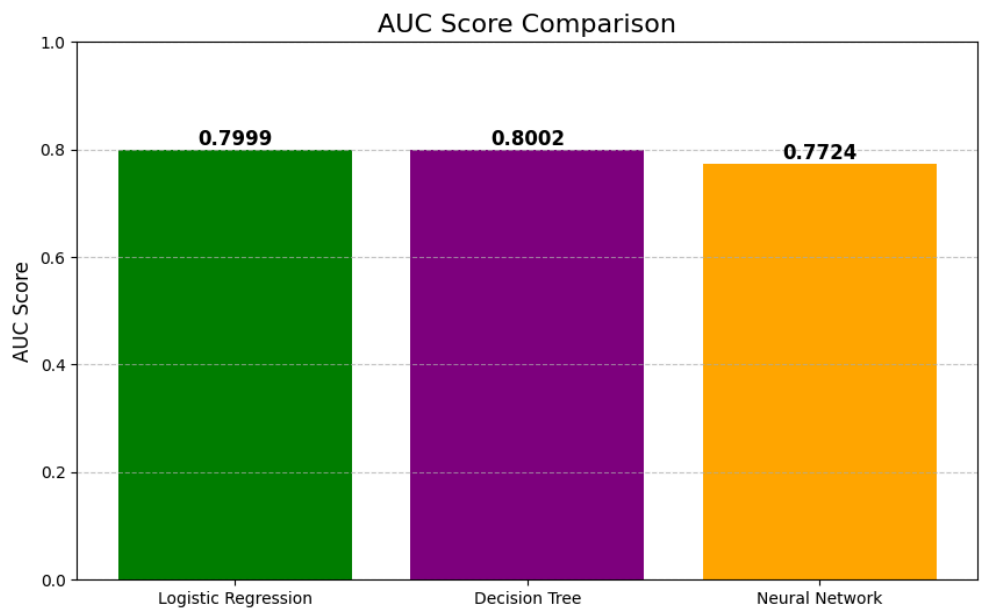




ROC Curve Comparison by Model



AUC Score Comparison by Model



Conclusion

Among the three models evaluated, Logistic Regression, Decision Tree, and a Neural Network (MLP) all demonstrated strong predictive performance, achieving accuracy rates significantly higher than random guessing (close to 90%). The Decision Tree emerged as the slight overall winner with the highest ROC-AUC score (0.8002) and Accuracy (89.98%), indicating high capability in distinguishing between subscribers and non-subscribers. However, the models exhibited distinct drawbacks in class wise performance due to the class imbalance.
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The consistently high ROC-AUC values (approximately 0.8) across all models suggest that the selected features offer strong discriminative power. However, the confusion matrix revealed that all models still struggle to identify the minority class (Yes), often favoring the majority class (No) to maximize accuracy.

Future improvements should focus on addressing this imbalance through techniques like SMOTE (oversampling) or Class Weights to boost the identification of potential subscribers. Overall, while the Neural Network offers the strongest statistical performance, the Decision Tree may be practically superior for campaigns prioritized on maximizing customer acquisition.